

Prevalence of HPV infection among females in the United States.

CONTEXT: Human papillomavirus (HPV) infection is estimated to be the most common sexually transmitted infection. Baseline population prevalence data for HPV infection in the United States before widespread availability of a prophylactic HPV vaccine would be useful. OBJECTIVE: To determine the prevalence of HPV among females in the United States. DESIGN, SETTING, AND PARTICIPANTS: The National Health and Nutrition Examination Survey (NHANES) uses a representative sample of the US noninstitutionalized civilian population. Females aged 14 to 59 years who were interviewed at home for NHANES 2003-2004 were examined in a mobile examination center and provided a self-collected vaginal swab specimen. Swabs were analyzed for HPV DNA by L1 consensus polymerase chain reaction followed by type-specific hybridization. Demographic and sexual behavior information was obtained from all participants. MAIN OUTCOME MEASURES: HPV prevalence by polymerase chain reaction. RESULTS: The overall HPV prevalence was 26.8% (95% confidence interval [CI], 23.3%-30.9%) among US females aged 14 to 59 years (n = 1921). HPV prevalence was 24.5% (95% CI, 19.6%-30.5%) among females aged 14 to 19 years, 44.8% (95% CI, 36.3%-55.3%) among women aged 20 to 24 years, 27.4% (95% CI, 21.9%-34.2%) among women aged 25 to 29 years, 27.5% (95% CI, 20.8%-36.4%) among women aged 30 to 39 years, 25.2% (95% CI, 19.7%-32.2%) among women aged 40 to 49 years, and 19.6% (95% CI, 14.3%-26.8%) among women aged 50 to 59 years. There was a statistically significant trend for increasing HPV prevalence with each year of age from 14 to 24 years ($P < .001$), followed by a gradual decline in prevalence through 59 years ($P = .06$). HPV vaccine types 6 and 11 (low-risk types) and 16 and 18 (high-risk types) were detected in 3.4% of female participants; HPV-6 was detected in 1.3% (95% CI, 0.8%-2.3%), HPV-11 in 0.1% (95% CI, 0.03%-0.3%), HPV-16 in 1.5% (95% CI, 0.9%-2.6%), and HPV-18 in 0.8% (95% CI, 0.4%-1.5%) of female participants. Independent risk factors for HPV detection were age, marital status, and increasing numbers of lifetime and recent sex partners. CONCLUSIONS: HPV is common among females in the United States. Our data indicate that the burden of prevalent HPV infection among females was greater

than previous estimates and was highest among those aged 20 to 24 years. However, the prevalence of HPV vaccine types was relatively low.